

Prompt Engineering — Quiz 1

Name: _____ Date: _____

Instructions: This quiz contains 20 multiple-choice questions. Answer all questions. Each question is worth 1 point unless otherwise noted. Total: 20 points.

#1 Which of the following best describes a **prompt**?

- A) A set of numerical parameters inside an LLM.
- B) The textual instruction, question, or example you give an AI system.
- C) The final output produced by an AI model.
- D) The hardware used to train a model.

#2 The simplest way to explain how most LLMs work is:

- A) They store exact answers to every possible question.
- B) They predict the next token (word piece) based on patterns learned from large text corpora.
- C) They run symbolic logic rules coded by engineers.
- D) They rely only on live internet access to answer questions.

#3 Tokenization can split a single long word into several sub-word tokens.

- A) True
- B) False
- C) Sometimes, but only for English
- D) Only for character-level tokenizers

#4 What does self-attention allow a Transformer model to do?

- A) Compare and weigh other tokens so each token's representation becomes context-aware.
- B) Store fixed dictionary definitions for words.
- C) Translate text into image features.
- D) Ignore long-range dependencies.

#5 Which statement about embeddings is correct?

- A) Embeddings are human-written definitions attached to each word.
- B) Embeddings are the final tokens produced by the model.
- C) Embeddings are numeric vectors that locate tokens in a “meaning space” so similar words are near each other.
- D) Embeddings are images generated by multimodal models.

#6 What does “hallucination” mean in the context of LLMs?

- A) A visual artifact in image synthesis.
- B) The model producing plausible-sounding but incorrect or fabricated information.
- C) A model refusing to answer due to safety filters.
- D) When a model repeats the input prompt verbatim.

#7 Increasing model size (more parameters) always guarantees fewer errors and no hallucinations.

- A) True
- B) False
- C) Only if trained on perfect data
- D) Only for language tasks

#8 Which of these is a useful prompt “lever” to improve AI output?

- A) Role (e.g., “You are a teacher”)
- B) Output format (e.g., “Return 3 bullets”)
- C) Audience (e.g., “for 9th graders”)
- D) All of the above

#9 In the self-attention mechanism each token computes Query, Key, and Value. Which description is most accurate?

- A) Query = what I’m looking for; Key = what I contain; Value = the information I provide to others.
- B) Query = the token index; Key = the token frequency; Value = the token length.
- C) Query = model parameters; Key = training data; Value = learning rate.
- D) All three are identical and interchangeable.

#10 Which of the following best describes “fine-tuning”?

- A) Training a model from scratch on random data.
- B) Additional training on curated examples to specialize or correct model behavior.
- C) Reducing the number of parameters to make the model smaller.
- D) Encrypting the model weights for security.

#11 What is an example of an emergent ability in large LLMs?

- A) Decreased compute requirements.
- B) Suddenly solving tasks with few examples (few-shot learning) not seen in smaller models.
- C) The model physically improving hardware.

D) Eliminating the need for tokenization.

#12 Why are position embeddings used in Transformers?

- A) To encode the order of tokens since attention is permutation-invariant.
- B) To compress tokens into fewer dimensions.
- C) To convert text into images.
- D) To increase the model temperature.

#13 Which of these is NOT a typical limitation of LLMs?

- A) Hallucination
- B) Biases inherited from training data
- C) Unlimited context-window (they can process arbitrarily long documents)
- D) Outdated knowledge due to training cutoff

#14 Tokenization helps models because:

- A) It allows handling rare or long words by splitting them into familiar subparts.
- B) It prevents models from learning anything new.
- C) It ensures outputs are always grammatically correct.
- D) It stores complete books inside tokens.

#15 Which is a correct example of a prompt that uses constraints effectively?

- A) "Write about photosynthesis."
- B) "You are a high-school biology teacher. Explain photosynthesis in 3 bullets for 9th graders in under 100 words."
- C) "Do anything you want."
- D) "Make me rich."

#16 Which metric describes how an embedding represents a token?

- A) A high-dimensional numeric vector where nearby vectors indicate semantic similarity.
- B) A single integer representing word length.
- C) A boolean flag marking stopwords.
- D) A 2D image file.

#17 What role does "instruction tuning" play in model behavior?

- A) It teaches the model to follow human instructions better via examples of prompts and desired outputs.
- B) It deletes the model's training data.
- C) It converts the model into a decision tree.
- D) It only changes the model's hardware.

#18 Which mitigation is effective against hallucinations?

- A) Verifying facts with reliable external sources or adding a retrieval/citation step.
- B) Asking the model to guess more often.
- C) Increasing temperature to the maximum.
- D) Removing tokenization.

#19 What is the purpose of multi-head attention in Transformers?

- A) It allows the model to attend to information from multiple representation subspaces and capture different relationships in parallel.
- B) It reduces the number of tokens by merging them.
- C) It encrypts the model weights for secure storage.
- D) It converts tokens into images for multimodal tasks.

#20 Which three elements do good prompts typically provide?

- A) Purpose, context, and constraints.
- B) Tokens, parameters, and gradients.
- C) Hardware, dataset, and compute budget.
- D) Output format only.

Scoring: Each question = 1 point. Total points: 20.